DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

Curriculum for B.Tech CSE V-Semester • Session 2026-27

Subject: Database Management Systems

Course Code: KCS501 • L-T-P: 3-1-0 • Credits: 4

UNIT I: Introduction & Entity-Relationship Model

- Overview of DBMS: Database System vs File System, Database Architecture, Three-Schema Architecture, Data Independence, Database Languages (DDL, DML)
- Entity-Relationship Model: ER Model Concepts, Entity Types, Relationship Sets, Constraints, Keys (Super Key, Candidate Key, Primary Key)
- ER Diagrams: Mapping Constraints, Extended ER Features (Generalization, Specialization, Aggregation), Reduction to Relational Schemas

UNIT II: Relational Data Model, Languages & SQL

- Relational Model: Relational Constraints (Domain, Key, Entity Integrity, Referential Integrity), Relational Algebra (Selection, Projection, Joins, Division)
- Relational Calculus: Tuple Relational Calculus, Domain Relational Calculus
- Structured Query Language (SQL): Data Types, DDL, DML, DCL, Views, Nested Sub-queries, Aggregate Functions, Joins
- PL/SQL Programming: Procedures, Functions, Cursors, Triggers

UNIT III: Database Design & Normalization

- Functional Dependencies: Definition, Trivial and Non-trivial FDs, Armstrong's Axioms, Closure of FD Sets, Minimal Cover
- Normal Forms: First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Boyce-Codd Normal Form (BCNF)
- Decomposition Properties: Lossless-Join Decomposition, Dependency Preservation
- Advanced Normalization: Multivalued Dependencies and 4NF, Join Dependencies and 5NF

UNIT IV: Transaction Processing & Recovery

- Transaction Concepts: Transaction States, ACID Properties, Concurrent Execution, Serializability (Conflict & View Serializability)
- Testing Serializability: Precedence Graphs, Recoverability, Cascadeless Schedules
- Failure and Recovery: Log-Based Recovery, Write-Ahead Logging (WAL), Checkpoints, Shadow Paging
- Distributed Databases: Distributed Data Storage, Directory Systems

UNIT V: Concurrency Control Techniques

- Concurrency Control Need: Lost Updates, Dirty Read, Unrepeatable Read
- Locking Protocols: Shared and Exclusive Locks, Two-Phase Locking (2PL), Strict 2PL, Rigorous 2PL
- Timestamp Protocols: Timestamp Ordering, Thomas Write Rule, Validation-Based Protocols
- Deadlock Handling: Wait-for Graph, Deadlock Prevention (Wait-Die, Wound-Wait), Deadlock Detection and Recovery

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University Examination Blueprint:

1. End Semester Exam Duration: 3 Hours • Maximum Marks: 100.
2. Minimum pass criteria: 40% in theory and 50% in sessional.
3. Calculators and electronic gadgets are strictly prohibited inside the examination hall.